PHOEBE CHEN

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EDUCATION

UNIVERSITY OF CALIFORNIA, LOS ANGELES — Expected Jun 2026

B.S. in **Statistics and Data Science** | Minor in **Data Science Engineering**

Relevant Coursework: Probability, Mathematical Statistics, Linear Models, Regression & Data Mining, Computational Statistics, Monte Carlo Methods, Neural Networks & Deep Learning, Optimization for Statistics, Causal Inference, Database Management Systems (IP), Machine Learning (IP), Bayesian Statistics (IP), Statistical Consulting (IP)

MOORPARK COMMUNITY COLLEGE — Aug 2022 - May 2024

Honors Program | Transfer to UCLA

PROJECTS

EMG-Based Keystroke Decoding with Neural Networks — Mar 2026

- Built and tuned a CNN–BiGRU deep learning model for EMG-based keystroke decoding, reducing character error rate to ~16% through architectural improvements and hyperparameter optimization (PyTorch)

Causal Analysis of Obesity and ICU Mortality — Mar 2026

- Performed data cleaning and feature engineering before estimating the causal effect of obesity on ICU mortality (MIMIC-IV) using logistic regression and IPTW, finding ~15% lower mortality odds among obese patients; evaluated robustness using alternative models/sensitivity checks and explored heterogeneity using causal forests

LAPD Crime Data Analysis Project — Mar 2026

- Processed 877K+ LAPD crime records using R (tidyverse), performing data cleaning, feature engineering, EDA, and statistical modeling (χ^2 tests, multinomial logistic, negative binomial) to analyze COVID-era crime patterns and create reproducible visualizations (Quarto, ggplot2, LaTeX)

Heart Failure Mortality Prediction — Dec 2025

- Conducted EDA on clinical data to identify key clinical predictors and feature importance; developed predictive models (logistic regression, random forest, neural network) to estimate mortality risk, achieving 81.3% validation accuracy and strong ROC-AUC performance

Graduate Admissions Equity Modeling — Dec 2025

- Developed predictive classification models (logistic regression, random forest, LASSO) to analyze MBA admissions and interpreted coefficients to understand demographic and academic drivers of outcomes; applied feature selection techniques (AIC/BIC, regularization), achieving 81.8% accuracy

UCLA ASA DataFest 2025 — Mar 2025

- Analyzed 190K+ records to identify geographic and industry-level business office migration trends

Median House Value Regression Model — Oct 2024

- Developed regression models on Census data and applied diagnostics and transformations to improve fit

EXPERIENCE

MOORPARK COLLEGE ENGINEERING CLUB — Sep 2022 – May 2024

Lead Programmer, Board Member — Secretary

- Led a 3-person programming team in qualifier competitions and the 2024 VEX U Robotics World Championship
- Coordinated software design, hardware–software integration, and debugging in time-constrained competition settings
- Developed autonomous navigation and control algorithms in C++ using PID controllers, improving autonomous task success rates by over 2x

SKILLS

Programming: Python, R, SQL, C++, MATLAB

Machine Learning/Statistics: Regression, Classification, Neural Networks, Model Evaluation, Hyperparameter Tuning, EDA

Tools: Git, Jupyter, RStudio, Visual Studio, LaTeX, Quarto, Figma